

CHIRAG MUKHIJA

Backend & Systems Engineer · B.E. Computer Science, BMS College of Engineering, Bengaluru

chiragmukhija.cs24@bmsce.ac.in | Bengaluru, India | chiragmukhija.vercel.app
github.com/Chirag-Mukhija | linkedin.com/in/chirag-mukhija | leetcode.com/u/c3CgTUeJ4W

SUMMARY

Backend-focused CS undergraduate (CGPA 9.2) who builds production-style systems from scratch — a distributed job scheduler, an Instagram-style backend and a payment API — with Node.js, PostgreSQL, Redis and Docker.

EDUCATION

B.M.S. College of Engineering, Bengaluru — B.E. in Computer Science and Engineering 2024 – 2028
CGPA: 9.2 / 10 · Coursework: Data Structures, Analysis of Algorithms, Operating Systems, OOP

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, SQL, C
Backend: Node.js, Express.js, REST APIs, WebSockets, Hono
Databases & Caching: PostgreSQL, Redis (sorted sets, Pub/Sub), MongoDB, SQLite
Infrastructure & Frontend: Docker, Docker Compose, Nginx, AWS S3, Git · React, Next.js, Tailwind CSS
Concepts: System design, job queues, idempotency, ACID transactions, feed fan-out, keyset pagination, rate limiting, DSA

PROJECTS

Distributed Job Scheduler — *Node.js, PostgreSQL, Redis, WebSockets, Docker* github.com/Chirag-Mukhija/Job-Scheduler

- Built a task queue from scratch (the model behind BullMQ and Celery) with an API service and worker processes that scale horizontally via `docker compose --scale worker=N`.
- Indexed due jobs in per-priority Redis sorted sets and claimed them with a guarded PostgreSQL UPDATE, removing full-table scans while ensuring no two workers run the same job.
- Added exponential-backoff retries, a dead-letter queue with manual requeue, priorities, delayed jobs, a live WebSocket dashboard over Redis Pub/Sub, and graceful SIGTERM shutdown that finishes in-flight jobs.

Instagram-Style Backend — *Node.js, PostgreSQL, Redis, S3, Nginx* github.com/Chirag-Mukhija/InstaBackend

- Implemented fan-out-on-write and fan-out-on-read feeds side by side to compare write amplification against read-time joins.
- Keyset pagination on all list endpoints and S3 presigned uploads: flat page cost at any depth, no image bytes on the server.
- Built JWT auth with Redis-backed revocable sessions and two-layer rate limiting (Nginx per-IP + Redis).

Payment Gateway API — *Node.js, Express 5, PostgreSQL (in progress)* github.com/Chirag-Mukhija/PaymentGateway

- Made payment creation idempotent with per-merchant idempotency keys; a unique-constraint backstop turns concurrent duplicates into a clean 409, not a double charge.
- One ACID transaction writes each payment with its first audit event; an append-only event log supports reconciliation, and amounts use DECIMAL to avoid floating-point drift.

ClipForge — Short-Video Pipeline — *TypeScript, SQLite, Remotion, ffmpeg, LLMs* github.com/Chirag-Mukhija/ShortsPipeline

- Turns a topic into a finished 1080×1920 explainer in about 5 minutes via a 10-step pipeline; each step is a durable SQLite job with retries and crash recovery.
- The LLM fills props for hand-built Remotion templates instead of writing animation code, so renders never break.

PRactice — Open-Source Practice Platform — *Next.js, FastAPI, GitHub API (team of 4)* github.com/sciboyrocks/PRactice

- Built the Issue → PR → Diff → File mapping that hands the LLM only the file a fix touched, not the whole repository.

ACHIEVEMENTS & ACTIVITIES

- Problem solving:** 200+ DSA problems across platforms; on LeetCode, 97 solved (61 Medium, 15 Hard), contest rating 1,460, 50 Days Badge 2026.
- Hackathons:** built Signal, a voice-to-tasks assistant (React Native, Express, Whisper, Gemini), at an AWS hackathon; competed in multiple college hackathons.
- Courses:** Fundamentals of Networking for Effective Backends (Hussein Nasser, Udemy); Sigma Full-Stack + DSA (Apna College); Fundamentals of Backend Engineering (in progress).
- Sports:** district-level silver medallist in swimming (100 m freestyle, 50 m backstroke); trains in the gym daily.